

Segmentation Generalization to a New Hospital

Can a prostate MRI model trained at one hospital work reliably at another?

PROJECT 4

Group D

Initial project presentation

Hospital MRI data differ

Scanner

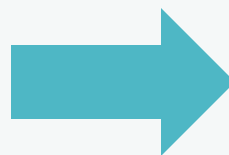
Manufacturer and hardware affect the measured signal

Acquisition protocol

Field strength, coil use, and scan settings vary

Image appearance

Contrast, resolution, and artifacts may change



A model may learn the training hospital's image style

When the data distribution changes, segmentation performance may decrease.

Same anatomy, different data distribution

Single-source domain generalization

TRAINING DOMAIN

Hospital A

T2 MRI + expert masks



Train and validate by patient



UNSEEN DOMAINS

Hospital B

External test only

Hospital C

External test only

Hospitals D-F

External test only

Does performance hold when the hospital changes?

No retraining or tuning on target hospitals

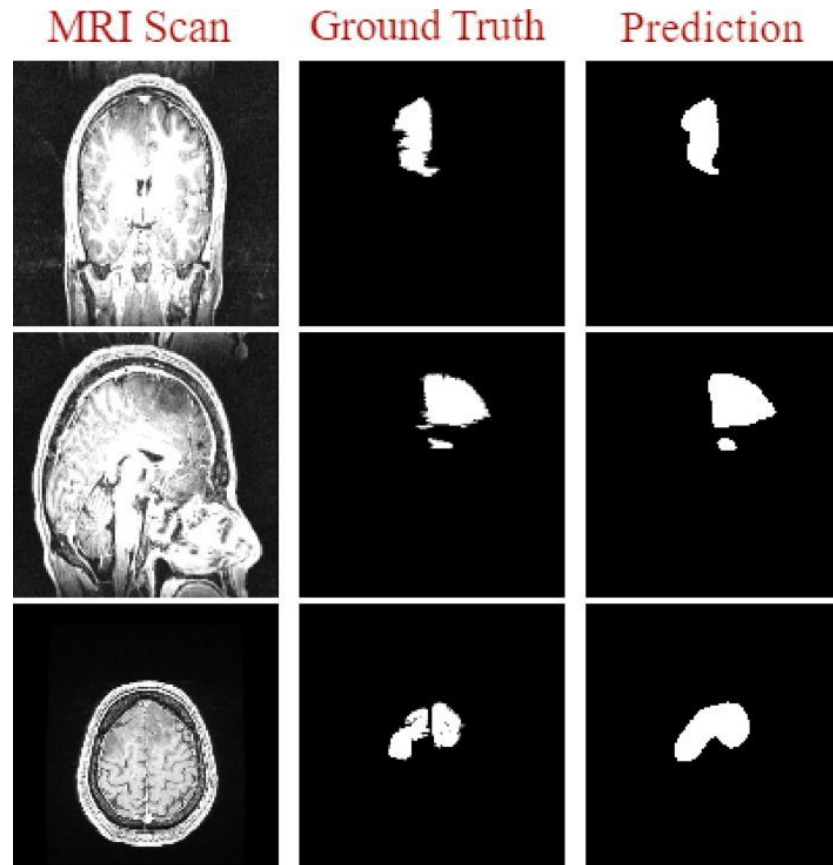
Six-site prostate MRI cohort

116 patients collected across institutions with different acquisition conditions

Site	Institution	Cases	Field (T)	Resolution, mm	Coil	Manufacturer
A	RUNMC	30	3	0.6-0.625 / 3.6-4	Surface	Siemens
B	BMC	30	1.5	0.4 / 3	Endorectal	Philips
C	HCRUDB	19	3	0.67-0.79 / 1.25	No	Siemens
D	UCL	13	1.5 and 3	0.325-0.625 / 3-3.6	No	Siemens
E	BIDMC	12	3	0.25 / 2.2-3	Endorectal	GE
F	HK	12	1.5	0.625 / 3.6	Endorectal	Siemens

Hospital identity is tied to scanner and acquisition differences

Prostate segmentation



Input

T2-weighted prostate MRI



Model

Pixel-level classifier

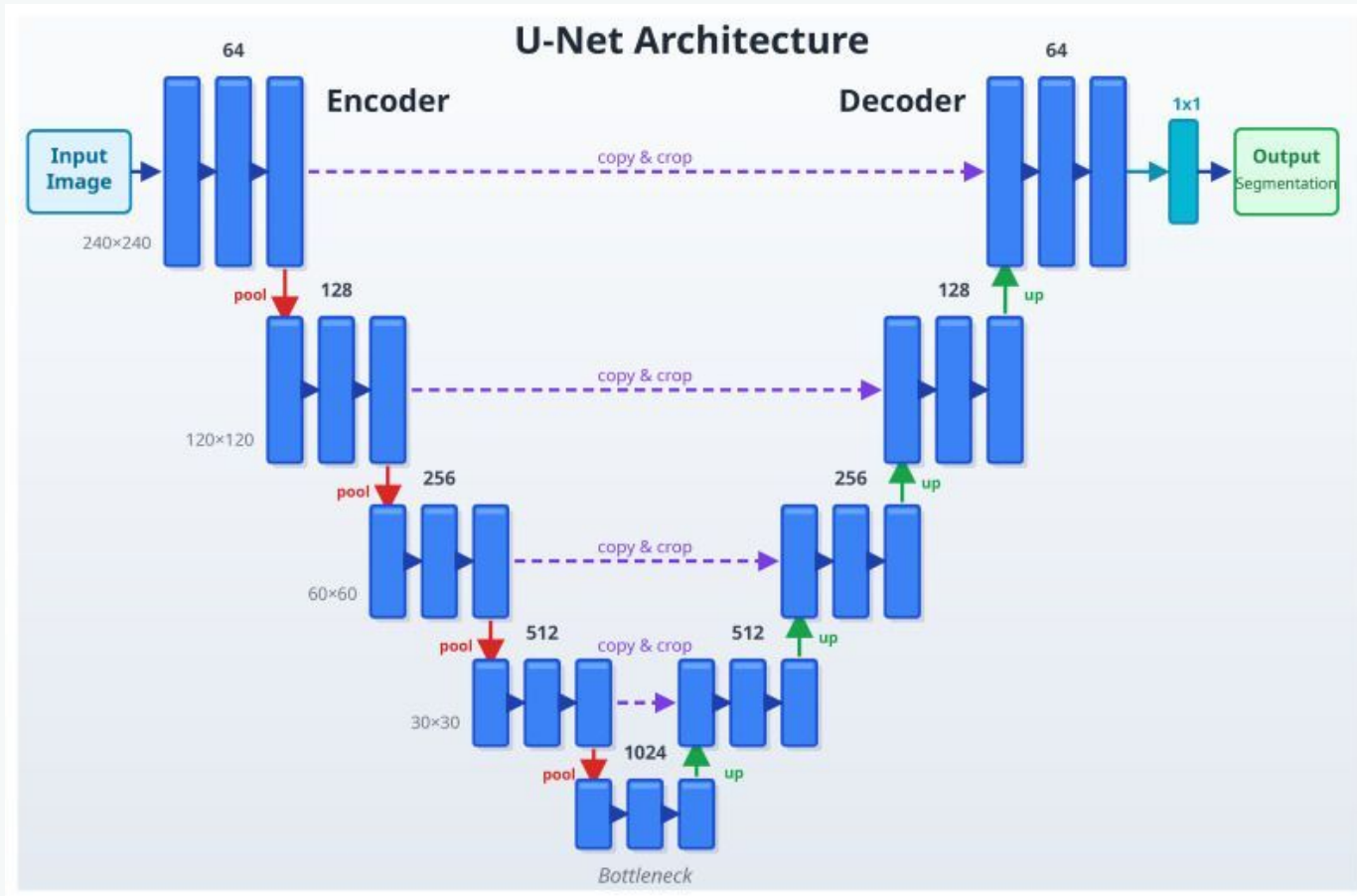


Output

Binary prostate mask

Each pixel: prostate or background

How U-Net creates a segmentation mask



Encoder

Compresses the image and learns features such as edges, texture, and shape

Bottleneck

Stores the most compressed, high-level representation

Decoder

Upsamples the features to predict a full-resolution mask

Skip connections restore spatial detail

SAML and our study

SAML, Liu et al. (2020)

The closest prior study uses the same six-site dataset.

SAML	Our initial study
Multiple source hospitals	One source hospital
One held-out target per run	Multiple unseen targets
Shape-aware meta-learning	Standard U-Net baseline
Tests a specialized DG method	Measures the generalization gap

SAML asks whether multi-source training transfers to one unseen hospital. We test how one locally trained model transfers across several hospitals.

Evaluation plan

1

Select the source site

Begin with RUNMC or another agreed source

2

Split by patient

Use source patients for training and validation

3

Train and freeze U-Net

No target-site tuning

4

Evaluate every target site

Report patient-level results by hospital

Evaluation logic

SOURCE

Hospital A

UNSEEN

Hospitals B-F

Compare the same metrics

Source Dice - target Dice

Stretch goal: repeat with each hospital as the source

Metrics and reporting

Dice score

Primary overlap metric. A score of 1 means perfect overlap.

ASD or HD95

Boundary error in millimeters. We will confirm which one the course prefers.

Generalization gap

Difference between source-site and external-site performance.

Report results at the patient level

For every hospital

Median, spread, and individual values

For every method

Use the same split and metrics

For failures

Show representative and worst cases

Do not hide site-specific failure inside one pooled average

First investigations and compute

First investigations

1

Dataset validation

Counts, image dimensions, spacing, orientation, and mask integrity

2

EDA by hospital

Representative images, intensity distributions, spacing, and prostate size

3

Baseline transfer

One source U-Net evaluated on every unseen hospital

Expected compute

Laptop

EDA, preprocessing, visualization, and small pipeline tests

One shared GPU

Full 2D U-Net training and initial evaluation

Cluster expansion

Repeated source sites, augmentation comparisons, and multiple seeds

Initial request: one V100-class GPU with 16-32 GB VRAM; no multi-GPU job required

Decisions before we begin training

01

Which hospital should be the first source?

02

Should every hospital eventually serve as the source?

03

Should our boundary metric be ASD or HD95?

04

Modern PyTorch baseline or direct SAML reproduction?

Can the model learn the anatomy instead of the hospital?